Shielded Q-learning for Thermally-Safe, Energy-Aware Multi-Model Intrusion Detection Scheduling on Lightweight Edge Devices

Nuonan Ouyang, Adrian Shatte, Zhigang Lu, Chao Chen, and Wei Xiang

***Abstract*—Edge intrusion detection requires detector selection to account for classification quality, processing deadlines, temperature, and energy. Feasible-Set Shielded Q-learning (FSSQL-R) combines a device-calibrated latency/thermal guard with a masked tabular policy. We evaluate its implementation on Raspberry Pi 3B+ and Pi 4B 8GB platforms, separating calibration, online deployment, and detector-generalization evidence. Independent held-out joint guard coverage is 2879/2880 on Pi 3B+ and 2877/2880 on Pi 4B. The Pi 3B+ guard excludes one detector under a binding workload, but FSSQL-R underperforms Safe-Greedy in F1 and reward. A corrected Pi 4B evaluation comprises 60 online runs, 36,000 windows, and 576,000 record evaluations, with direct DC input-energy measurement. FSSQL-R and Safe-Greedy have similar mean F1 under no and moderate background load (approximately 0.960), whereas under the highest tested load their mean F1 is 0.8869 and 0.9608, respectively. A post hoc trace analysis links 1322/3000 high-load FSSQL-R windows to unvisited training states, zero-valued Q rows, and the fixed tie-breaking action. Neither the learning layer nor the measured energy results establish a consistent advantage over simpler controllers. The findings distinguish calibrated action admissibility from end-to-end safety and show why state support, false positives, controller cost, and physical energy must accompany average detection scores in evaluations of learning-based IDS schedulers.**

***Index Terms*—Intrusion detection, edge computing, constrained scheduling, safe reinforcement learning, Q-learning, runtime shielding, thermal constraints, energy measurement.**

I. Introduction

Lightweight edge devices increasingly execute intrusion detection close to data sources, but deployment is constrained by latency, thermal headroom, and energy. A detector that performs well offline may be unsuitable for a particular device or workload. Keeping several detectors available and selecting among them online offers an alternative to always executing a single model [1]–[3]. The scheduler itself, however, consumes time and introduces another source of error.

Reinforcement learning (RL) can optimize sequences of decisions, but unconstrained exploration may exceed operating limits. Constrained MDP approaches typically limit expected costs [4]–[6]. Runtime shielding instead removes actions that fail an admissibility test before the policy acts [7]–[9]. Safety-aware RL has also been applied to IDS remediation in 5G-enabled IoT [10]. These approaches motivate separating a learned objective from a device-specific feasibility check; they do not imply that learning will outperform a simpler controller.

We study Feasible-Set Shielded Q-learning (FSSQL-R) for choosing among four resident IDS models under 40-ms modeled-latency and 82°C operating caps. The questions are whether the guard restricts model choice when resources are constrained, whether Q-learning adds value beyond a same-guard greedy rule, and what controller and energy costs are incurred in online execution. An independently calibrated Pi 4B implementation provides a second hardware configuration and a conventional temperature/queue/risk threshold baseline.

The evidence does not support a general advantage for Q-learning. Safe-Greedy performs better in the Pi 3B+ stationary binding experiment. In the corrected Pi 4B experiment, the learned and greedy policies have similar detection scores under lower loads, but the learned policy degrades under the highest background load. Direct energy measurements also fail to show consistent savings. The contribution is a measured account of these boundaries:

1) A device-specific feasible-action formulation with separately fitted predictors, residual margins, and held-out coverage, together with an explicitly conditional safety statement.
2) Matched-seed comparisons against greedy, fixed-model, and threshold controllers, including FPR, per-seed outcomes, and a trace-level analysis of unsupported Q states.
3) Online timing on two Pi configurations, and direct DC input-energy measurements across three Pi 4B background-load settings, with explicit accounting boundaries.
4) Separate reporting of scheduler-development workload performance and the saved fixed-threshold evaluation of four detectors on all 82,332 official UNSW-NB15 test rows.

II. Related Work and Positioning

A. Sustainable scheduling and adaptive edge inference

Recent work in sustainable computing increasingly treats resource management as an online control problem rather than a static configuration task. In *IEEE Transactions on Sustainable Computing* (TSUSC), Wang *et al.* use deep reinforcement learning (DRL) for runtime GPU frequency capping [11]; Ko *et al.* schedule energy-harvesting IoT sleep states while preserving reconstruction quality [12]; Gao *et al.* jointly optimize latency and energy for in-network task scheduling [13]; and Li *et*

N. Ouyang (corresponding author) and A. Shatte are with the College of Science and Engineering, James Cook University, Australia (e-mail: nuonan.ouyang@my.jcu.edu.au; adrian.shatte@jcu.edu.au).

Z. Lu is with the School of Computer, Data and Mathematical Sciences, Western Sydney University, Australia (e-mail: z.lu@westernsydney.edu.au).

C. Chen is with the School of Accounting, Information Systems & Supply Chain, RMIT University, Australia (e-mail: chao.chen@rmit.edu.au).

W. Xiang is with the Australian Centre for AI in Medical Innovation and the Cisco–La Trobe Centre for AI and Internet of Things, La Trobe University, Australia (e-mail: w.xiang@latrobe.edu.au).

al. learn DVFS policies for multitask edge workloads under deadlines [14]. Recent TSUSC work also highlights adaptation in IoT anomaly detection under concept drift [15]. These studies motivate dynamic resource-aware control, but their actions are frequency, sleep, task-placement, or adaptation decisions rather than selection among heterogeneous on-device IDS models.

Closely related edge-inference work in *IEEE Transactions on Mobile Computing* (TMC) reinforces the need to account for workload variation and the cost of the controller itself. Tan *et al.* formulate workload-adaptive edge inference request scheduling as an MDP and use RL to select among model-serving choices [16]; Zhang *et al.* co-optimize DVFS and edge-cloud offloading for latency/energy efficiency [17]; and thermal-aware scheduling has been studied explicitly for mobile DNN execution [3]. Our problem differs in two respects: model selection is constrained by a separately calibrated per-window feasible set, and our evaluation measures instrumented online controller latency and physical input energy on the target device rather than relying only on a modeled cost.

B. Lightweight and online intrusion detection

Lightweight IDS research spans classical one-class methods, extreme learning machines, compact deep models, and TinyML-style deployment [18]–[23]. More recent high-end security literature has emphasized adaptation as well as compact execution. Nakip and Gelenbe propose an online self-supervised IDS that adapts without conventional offline relabeling [24], while Yang *et al.* address lightweight dynamic open-set intrusion detection for IIoT [25]. These works focus on improving or updating the detector. The practical problem here is complementary: the detectors are frozen before formal evaluation, and the runtime controller decides which detector to execute under latency, thermal, and resource constraints.

C. Safe RL and shielding

Safe RL covers expectation-constrained optimization, Lyapunov methods, and action shielding [5]–[7], [26]. FSSQL-R is not novel because shielding itself is new; rather, it instantiates a calibrated, device-specific feasible-action layer for multi-model IDS scheduling and separates this layer from utility optimization. Malwi *et al.* [10] use a runtime safety shield for self-healing remediation actions in 5G-IoT. Our action semantics and constraints are different: the actions are heterogeneous IDS models, and admissibility is derived from device-specific latency and thermal measurements under explicit operating caps.

D. Positioning against simpler controllers

A key lesson from the revision is that learning must be compared with controllers that do not require training. The baseline set therefore includes Static-Light, Round-Robin, Safe-Greedy, Unshielded-Q, and a conventional threshold controller. Safe-Greedy uses the same calibrated feasible set as FSSQL-R but makes a one-step security-utility choice. Unshielded-Q uses the same state, reward, and Q-learning implementation without feasible-set masking. The threshold controller uses only measured state and fixed development-time thresholds. Earlier energy-aware and multi-objective precursor formulations are not archival publications and are not used to establish novelty. The numeric Q-learning configuration used here is reported explicitly, fixed before formal evaluation, and evaluated against these simpler alternatives.

III. Problem Formulation

Decision windows have duration $\Delta = 100$ ms. Action $a_t \in \mathcal{A}$ selects one of four resident revision implementations: FISVDD, LUCID, TinyDL, and OI-SVDD+AS-ELM. The state is

$$s_t = (z_t, T_t, b_t, q_t, a_{t-1}), \tag{1}$$

where z_t is estimated traffic risk, T_t is measured CPU temperature, b_t is a controller-internal energy-state proxy, q_t is queued backlog, and a_{t-1} is the previous action. The energy state is not a measured battery state. In the Pi 4B online experiment $q_t = 0$ throughout; no queue-recovery claim is made.

The label-free one-step reward is

$$\begin{aligned} r_t = {} & z_t \,\mathrm{TPR}_{\mathrm{val}}(a_t) - (1 - z_t)\,\mathrm{FPR}_{\mathrm{val}}(a_t) \\ & - 0.1 e(a_t) - 0.05 \min(q_t/256, 1) \\ & - 0.02 \mathbb{I}[a_t \neq a_{t-1}], \end{aligned} \tag{2}$$

with no switching penalty on the first action. The fixed detector-quality terms come from validation data. $e(a)$ is a normalized compute/energy proxy, not measured joules. Its Pi 4B values are 0.352513, 1.000000, 0.426251, and 0.584404 in the action order above and are unchanged across B0–B2. The energy state starts at 1 and decreases by $0.001e(a_t)$ per window. Ground-truth evaluation labels do not enter this reward.

Let $U_\tau(s,a) = \widehat{\tau}(s,a) + \epsilon_{\tau,a} + \delta_\tau$ and $U_T(s,a) = \widehat{T}_{t+1}(s,a) + \epsilon_T + \delta_T$ be upper predictions with one-sided residual margins and fixed buffers. The admissible set is

$$\mathcal{A}_{\mathrm{safe}}(s) = \{a : U_\tau(s,a) \leq 40 \text{ ms},\ U_T(s,a) \leq 82^\circ\mathrm{C}\}. \tag{3}$$

The guarded quantity is the calibrated action/service measurement, not an unconditional bound on every operation in the runtime. When the set is empty, the implementation flags fallback and selects the fixed FISVDD action; that fallback is not guaranteed safe. No accepted online run used it.

IV. Feasible-Set Shielded Q-learning

A. Device-specific calibration

Latency prediction is an empirical fit-split 0.99-quantile lookup indexed by action, batch size, and background setting. Temperature uses a standardized one-step linear model with eight design columns. Separate fit and margin windows determine the predictor and positive-residual 0.99 quantiles. Fixed buffers of 10 ms and 0.6°C are added. These quantities are frozen before held-out coverage is evaluated; the supplement specifies residual distributions and device-specific margins.

TABLE I
RECENT WORK MOST CLOSELY RELATED TO SUSTAINABLE SCHEDULING, EDGE INFERENCE, AND ADAPTIVE IDS.

Work	Venue	Runtime decision	Main evidence/constraint	Relation to this work
Wang *et al.* [11]	TSUSC 2024	GPU frequency cap	DRL-based runtime power control	Sustainable runtime control; no IDS feasible-set guard
Xu *et al.* [15]	TSUSC 2024	Drift adaptation	IoT anomaly adaptation across datasets	Adaptive security model rather than detector scheduling
Ko *et al.* [12]	TSUSC 2025	IoT sleep scheduling	Energy harvesting and outage-aware DRL	Energy-aware IoT control; different action and constraint semantics
Gao *et al.* [13]	TSUSC 2025	In-network task scheduling	Joint latency/energy optimization	Related multi-objective scheduling, primarily network/compute placement
Li *et al.* [14]	TSUSC 2025	DVFS selection	Deadline-aware edge execution	Device adaptation through frequency, not IDS model selection
Tan *et al.* [16]	TMC 2024	Edge inference requests/models	Workload-adaptive RL scheduling	Closest adaptive inference framing; no calibrated IDS shield
Zhang *et al.* [17]	TMC 2024	DVFS + offloading	Physical heterogeneous edge evaluation	Latency/energy control through DVFS/offloading
Nakip and Gelenbe [24]; Yang *et al.* [25]	TIFS 2024–25	Online/open-set IDS adaptation	Dynamic IDS learning and lightweight execution	Improves detector adaptation; our focus is constrained runtime selection among frozen detectors
This work	TSUSC	IDS model selection	Device-calibrated latency/thermal guard, bounded online timing, direct DC input energy	Higher-level complement to DVFS/power control; no consistent energy-saving claim

B. Masked updates and evaluation

For nonterminal transitions the update is

$$Q(s_t,a_t) \leftarrow Q(s_t,a_t) + \alpha_t\big[r_t + \gamma \max_{a' \in \mathcal{A}_{\text{safe}}(s_{t+1})} Q(s_{t+1},a') - Q(s_t,a_t)\big]. \quad (4)$$

Four bins each for risk, temperature, energy, and backlog, and five previous-action values, produce 1280 states and 5120 state-action pairs. The bins, learning schedules, and replay settings are in Table II. The corrected Pi 4B policy is trained for 6000 windows per seed using 16-record batches, the Pi 4B thermal transition model, and B0 development data. Formal B1/B2 evaluation does not retrain it. Q values start at zero. Evaluation uses $\epsilon = 0$ with no Q or replay updates; a tie selects the lowest admissible action index. This also defines behavior in previously unvisited states.

C. Conditional admissibility

Proposition 1. *If the one-step positive latency and temperature prediction errors do not exceed their respective residual margins plus fixed buffers, an admitted action satisfies the modeled caps in* (3)*.*

Proof. For an admitted action, $U_\tau \le 40$ ms and $U_T \le 82°$C. Under the stated error bounds, the corresponding realized quantities do not exceed these upper predictions. □ □

Empirical quantiles do not establish these bounds for all future states. Coverage, observed cap exceedances, and online timing are therefore reported separately. Zero observed violations are not a proof of zero risk.

V. EXPERIMENTAL METHOD

A. Reconstruction, correction, and test chronology

The submitted results used hardware-profiled replay. Because the original execution artifacts were unavailable, the revision reconstructs and freezes the detectors, preprocessing, calibration, and scheduler. These are not byte-identical historical artifacts, and the new tables replace rather than numerically reproduce the earlier tables. The Pi 3B+ online configuration also uses a newly documented stabilized power and cooling arrangement.

An implementation audit subsequently found that the earlier Pi 4B online worker did not implement the declared batch size, risk/controller logic, reward, and background-load behavior consistently; timing and energy alignment were also affected. Those Pi 4B results are excluded. All Pi 4B numbers here come from the corrected R6 campaign, which uses the actual 16-record inference path, configuration-driven thresholds, masked selection, and contemporaneous load-process records. Earlier attempts remain in the artifact history but are not pooled with R6.

UNSW-NB15 official training data [27] provide group-separated model-training, validation, and scheduler-development partitions. Validation TPR/FPR alone supplies detector-quality terms to the reward. The submitted wording that test-set statistics supplied these terms describes a leakage path; unavailable historical code cannot establish that it was only a wording error. That path is absent from the revision implementation.

The official detector test was evaluated once on 11 September 2026. The October implementation corrections and R6 runs occurred afterwards, using frozen development inputs without reopening the test. The four detectors, preprocessor, thresholds, and validation-quality file retain the hashes recorded at the earlier test evaluation. Thus the saved test results concern those fixed detectors; the later scheduler evaluation is not an untouched-test evaluation. The preserved test aggregates allow confusion-matrix recomputation, but do not supply a complete historical prediction/copy lineage for independently regenerating all ranking metrics.

B. Hardware and guard evidence

The Pi 3B+ uses a heatsink, continuously running fan, and open-top case; the Pi 4B 8GB uses active cooling. The governor is fixed to `performance`. Hardware faults and departures from the specified starting conditions invalidate an attempt, independently of its scientific outcome. Each platform has separate fit, margin, and coverage splits of 2880 windows:

TABLE II
PARAMETERS USED IN THE RECONSTRUCTED EVALUATION.

Parameter	Value
Decision interval; caps	100 ms; 40 ms and 82°C
Guard cells/split	$4 \times 6 \times 3 \times 40 = 2880$
Guard batch sizes	1, 16, 64, 256, 512, 1024
Risk/energy bin edges	0.25, 0.50, 0.75
Temperature bin edges	35, 45, 55°C
Backlog bin edges	0.5, 2, 8 records
States/state-action pairs	1280/5120
Discount; learning rate	$\gamma = 0.95$; $\alpha = \min(0.5, (1+N)^{-0.6})$
Exploration	Linear 1 to 0.05 by window 4500
Replay	FIFO 512; 1 online + 4 replay updates
Pi 4B Q training	5 seeds × 6000 windows, n16/B0
Pi 4B evaluation	4 policies × 3 loads × 5 seeds; 600 windows/run
B0/B1/B2	No worker / one CPU at configured 50% / 100% duty
Meter	KM003C; nominal 10 samples/s, DC input

TABLE III
INDEPENDENT HELD-OUT EMPIRICAL COVERAGE, 2880 WINDOWS/DEVICE.

Device	Latency	Thermal	Joint
Pi 3B+	2879/2880	2880/2880	0.999653
Pi 4B 8GB	2880/2880	2877/2880	0.998958

four actions, six batch sizes, three background settings, and 40 observations per cell. Predictor parameters and margins are not transferred between devices.

C. Controllers and workloads

Safe-Greedy maximizes $z_t \mathrm{TPR}_{\mathrm{val}}(a) - (1 - z_t)\mathrm{FPR}_{\mathrm{val}}(a)$ over the same admissible set as FSSQL-R. It is a same-guard security-greedy reference, not a full-objective myopic ablation: the comparison does not isolate decision horizon alone. Unshielded-Q, evaluated on Pi 3B+, removes the masking from the reconstructed Q implementation; it is not an exact reproduction of an earlier author scheduler. Round-Robin cycles through all models. Static-Light always selects the preregistered light action, FISVDD, including its poor detection performance.

The Pi 4B Threshold controller is a no-training, unshielded baseline with hysteresis. It selects FISVDD when $T \geq 80.1495$°C or $n + q \geq 512$; it selects TinyDL when $T \leq 78.1495$°C, $n + q \leq 256$, and $z \geq 0.7$; otherwise it retains the previous action, initially FISVDD. These are configured thresholds, not values inferred from test outcomes. The temperature/queue branches are implemented but are not activated in the evaluated cold, zero-backlog regime; their implementation is checked separately.

The Pi 3B+ binding comparison contains six policies, ten matched seeds, and 600 windows/run. Pi 4B compares FSSQL-R, Safe-Greedy, Threshold, and Static-Light at n16/B0, n16/B1, and n16/B2, using seeds 5101–5105. For each seed a permutation of the 6000 development records defines $i_{w,j} = \pi[(16w + j) \bmod 6000]$. The same 600-window schedule is used across policies and loads. This gives 9600 record evaluations/run and 576,000 across the accepted matrix, not that many independent observations. Risk is the mean of 16 frozen RiskProxy outputs. Labels are stored separately and loaded only after the decision loop for scoring.

B1/B2 start a separate worker pinned to CPU 3 at configured 0.5/1.0 duty in a 100-ms period. PID, affinity, duty, and process liveness are recorded during each run. No independent process-utilization time series is available, so these values describe the programmed load rather than a measured CPU percentage. Run order is fixed within seed blocks; cooldown uses the documented idle-temperature criterion. R6 retains 58 original valid cells and two replacements, cells 21 and 45, whose initial temperature checks failed. Both original attempts remain archived; replacement metrics are never mixed with the earlier attempts.

D. Timing and physical energy

Models remain resident. Pi monotonic timestamps measure preprocessing, risk estimation, telemetry, state construction, latency/thermal guards, admissible-set construction, policy selection, action binding, and inference. "Controller" runs from after preprocessing through action binding. E2E runs from preprocessing start through post-action telemetry; reward calculation, row serialization, and end-of-run label scoring are outside that E2E boundary. Switching timing measures resident action binding, not model reload. Fixed 100-ms scheduling targets are retained, including jitter and overruns.

KM003C sampling is performed on the Mac outside the Pi decision path. Actual sample timestamps are integrated by the trapezoidal rule between Mac start/end ACK receipts, with at least 5 s of surrounding samples and no endpoint extrapolation. This is absolute DC input energy without idle subtraction, not AC wall energy. The acknowledged interval includes scheduled idle periods, background load, and end-of-run scoring/writeout; its observed duration is 60.070–60.282 s. $E/600$ is run energy normalized per window, not a resolved 100-ms energy trace. No measured joules are substituted into the frozen proxy reward.

E. Statistical analysis

The unit is the matched run/seed, not a window or repeated record. We report all seeds and use means/SDs for continuous summaries and median/range for FPR. Paired differences use a two-sided exact sign-flip test and 2000 paired percentile-bootstrap resamples (seed 20261022). At five pairs the smallest two-sided sign-flip p is 0.0625; intervals and tests therefore need not yield identical decisions. Multiple endpoints and policy contrasts are descriptive, with unadjusted p values reported in the artifact, not a family of confirmatory superiority claims. All five Pi 4B seeds are retained. State-support diagnostics are explicitly post hoc and do not trigger retraining.

VI. RESULTS

A. RQ1: Guard coverage and action admissibility

Table III retains every held-out miss: one on Pi 3B+ and three thermal-bound misses on Pi 4B. Exceeding a thermal prediction bound is not the same as exceeding the 82°C operating cap. The Pi 3B+ binding experiment has three admissible actions throughout, excluding LUCID; the Pi 4B R6 experiment has four throughout. Hardware and workload both differ, so this

TABLE IV
PI 3B+ BINDING-RESOURCE RESULTS: MEANS OVER TEN MATCHED SEEDS. FULL DISPERSION AND PAIRED VALUES ARE IN THE SUPPLEMENT AND ARTIFACT.

Policy	F1	Reward	E2E ms	Switch cost	Viol.
FSSQL-R	0.8523	0.5434	8.3149	5.348	0
Safe-Greedy	0.8708	0.5839	7.9775	0	0
Threshold	0.8704	0.5834	7.9704	0.012	1
Unshielded-Q	0.8523	0.5489	10.1141	4.438	0
Round-Robin	0.8148	0.4801	10.6571	11.980	0
Static-Light	0.7203	0.3389	8.2550	0	0

contrast is not a controlled estimate of processor-generation effects.

Pi 4B records no observed latency/thermal cap violation or fallback. Its maximum E2E latency is 18.375 ms, maximum recorded temperature is 47.712°C, and minimum recorded CPU frequency is 1800 MHz. These conditions do not establish the effectiveness of binding thermal control. The Pi 3B+ Threshold run contains one retained 52.522-ms E2E event, dominated by a 50.414-ms controller interval while inference takes 0.446 ms. For that action, the point prediction is 5.147 ms and the guarded upper prediction is 15.703 ms, leaving 24.297 ms to the cap. The cause of the controller tail is not resolved by these timings; the observation limits an unconditional complete-pipeline safety claim.

B. RQ2: Learned and deterministic controllers

The Pi 3B+ comparison in Table IV favors Safe-Greedy. Exact paired sign-flip tests across ten seeds give $p = 0.00195$ for F1, reward, and switching cost, and $p = 0.00586$ for mean E2E latency. Both FSSQL-R and Safe-Greedy have zero observed violations, so no safety advantage is inferred from their comparison.

The corrected Pi 4B results do not show the previously reported advantage over Safe-Greedy. Under B0/B1, FSSQL-R mean F1 is 0.9598, compared with 0.9608 for Safe-Greedy. Under B2, its mean F1 falls to 0.8869 and mean FPR rises to 0.5199, while Safe-Greedy remains at F1 0.9608 and FPR 0.1327. The B2 paired F1 difference is -0.0739 with bootstrap interval $[-0.0888, -0.0568]$ and exact $p = 0.0625$. The FPR difference is $+0.3872$, interval $[+0.2933, +0.4740]$, also $p = 0.0625$. These substantial directional differences should not be reported as a two-sided $p < 0.05$ result.

Safe-Greedy chooses TinyDL in every Pi 4B window; it is not an all-positive controller in R6. Threshold attains mean F1 0.9605 and FPR 0.1341 in each condition, close to Safe-Greedy. Static-Light retains FISVDD and has FPR 1.0. The simplicity of the former two rules is a practical reference against which learning must be justified, not a baseline deficiency to be removed.

C. Training support and the high-load result

A post hoc lookup of each runtime state in the frozen Pi 4B training visits identifies 0/3000 unvisited-state windows in B0, 1/3000 in B1, and 1322/3000 in B2. Every one of the 1322 B2 windows has an all-zero Q row and selects action 0, FISVDD, under the fixed lowest-index tie rule. This is a directly reproducible execution path, not an estimate of failure probability or proof of a dynamical attractor. Temperature, energy-state bins, and previous-action histories determine which states are reached; no controlled ablation separates their causal contributions here.

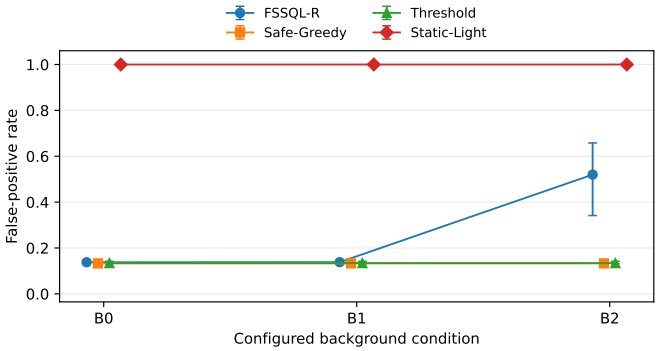


Fig. 1. Corrected Pi 4B FPR. Points are policy means and error bars span the five observed seed values; all seed-level values are retained in the supplement. The high-load deterioration concerns FSSQL-R, not Safe-Greedy.

The Pi 4B training checkpoints give mean rolling reward 0.4736 at 600 windows and 0.5541 at 6000, with 22.2/73.4 and 25.8/91.0 visited states/state-action pairs, respectively. The final mean greedy-policy change rate is zero over the checkpoint comparison, but mean absolute TD error remains 0.1293 and most states are unvisited. Pi 3B+ has similarly sparse support (34.9/118.9 at 6000 windows). These are finite-budget stability diagnostics on visited states, not evidence of full-state convergence. The separate preregistered Pi 3B+ queue-transition extension failed its accumulation/recovery preflight and produced no policy results.

D. RQ3: Controller and physical-energy costs

Across B0–B2, mean controller time is 4.673 ms for FSSQL-R and 4.516 ms for Safe-Greedy, compared with selected-detector inference of 0.520 and 0.441 ms. E2E means are 8.762 and 8.507 ms under the stated boundary. Controller work is thus material, but the timer excludes reward/serialization/scoring and should not be interpreted as all software overhead. The mean FSSQL-R switch count rises from 22.6/23.0 in B0/B1 to 162.8 in B2; Safe-Greedy has no switches. The proxy switching penalty and measured action-binding time are different quantities.

The paired FSSQL-R-minus-Safe-Greedy energy differences are $+0.908$ J (B0), -1.209 J (B1), and $+0.645$ J (B2). The direction is not consistent across loads; exact paired p values are 0.125, 0.125, and 0.5625. FSSQL-R mean DC input power rises from 2.948 W to 3.325 W and 3.733 W across the three configured loads. These measurements describe operation of the Pi, controller, resident detectors, and load worker over the acknowledged interval, not energy attributable to an individual inference. They do not support a general energy-saving claim.

TABLE V
CORRECTED PI 4B ONLINE EVALUATION: FIVE MATCHED SEEDS PER CONDITION, 600 WINDOWS OF 16 RECORDS PER RUN. FPR IS MEDIAN [MINIMUM, MAXIMUM]; OTHER ENTRIES ARE MEAN ± SD. ALL FOUR POLICIES HAVE ZERO OBSERVED LATENCY AND THERMAL VIOLATIONS. ENERGY INCLUDES THE ACKNOWLEDGED RUN INTERVAL, NOT DETECTOR EXECUTION ALONE.

Load	Policy	F1	FPR: median [range]	Mean reward	P99 E2E (ms)	Controller (ms)	Energy (J)
B0	FSSQL-R	0.9598 ± 0.0009	0.1377 [0.1302,0.1453]	0.5803 ± 0.0038	11.279 ± 1.247	4.660 ± 0.011	177.39 ± 0.94
B0	Safe-Greedy	0.9608 ± 0.0008	0.1309 [0.1302,0.1386]	0.5832 ± 0.0022	10.081 ± 0.083	4.497 ± 0.013	176.48 ± 0.53
B0	Threshold	0.9605 ± 0.0010	0.1328 [0.1309,0.1418]	0.5827 ± 0.0022	10.036 ± 0.279	4.458 ± 0.019	176.02 ± 0.93
B0	Static-Light	0.8101 ± 0.0011	1.0000 [1.0000,1.0000]	0.3341 ± 0.0039	10.322 ± 0.221	4.462 ± 0.023	176.64 ± 0.78
B1	FSSQL-R	0.9598 ± 0.0008	0.1377 [0.1315,0.1453]	0.5802 ± 0.0038	11.157 ± 0.994	4.687 ± 0.019	200.07 ± 1.21
B1	Safe-Greedy	0.9608 ± 0.0008	0.1309 [0.1302,0.1386]	0.5832 ± 0.0022	10.389 ± 0.375	4.531 ± 0.020	201.28 ± 0.86
B1	Threshold	0.9605 ± 0.0010	0.1328 [0.1309,0.1418]	0.5827 ± 0.0022	10.478 ± 0.438	4.503 ± 0.021	200.97 ± 2.09
B1	Static-Light	0.8101 ± 0.0011	1.0000 [1.0000,1.0000]	0.3341 ± 0.0039	10.239 ± 0.141	4.454 ± 0.011	200.76 ± 1.18
B2	FSSQL-R	0.8869 ± 0.0203	0.5425 [0.3414,0.6579]	0.4671 ± 0.0327	11.374 ± 0.791	4.672 ± 0.006	224.60 ± 2.08
B2	Safe-Greedy	0.9608 ± 0.0008	0.1309 [0.1302,0.1386]	0.5832 ± 0.0022	10.487 ± 0.290	4.521 ± 0.027	223.95 ± 1.42
B2	Threshold	0.9605 ± 0.0010	0.1328 [0.1309,0.1418]	0.5827 ± 0.0022	10.361 ± 0.488	4.499 ± 0.018	224.09 ± 1.89
B2	Static-Light	0.8101 ± 0.0011	1.0000 [1.0000,1.0000]	0.3341 ± 0.0039	10.306 ± 0.231	4.441 ± 0.013	223.31 ± 0.84

TABLE VI
FSSQL-R MINUS SAFE-GREEDY ON PI 4B. PAIRED MEAN DIFFERENCE AND PERCENTILE BOOTSTRAP 95% INTERVAL; EXACT TWO-SIDED SIGN-FLIP p. FIVE PAIRS PER CONDITION.

Load	Metric	Difference	95% interval	p
B0	F1	-0.0009	[-0.0019,-0.0002]	0.1250
B0	FPR	+0.0051	[+0.0011,+0.0103]	0.1250
B0	Energy (J)	+0.908	[+0.173,+1.867]	0.1250
B1	F1	-0.0010	[-0.0019,-0.0003]	0.0625
B1	FPR	+0.0054	[+0.0016,+0.0103]	0.0625
B1	Energy (J)	-1.209	[-1.884,-0.440]	0.1250
B2	F1	-0.0739	[-0.0888,-0.0568]	0.0625
B2	FPR	+0.3872	[+0.2933,+0.4740]	0.0625
B2	Energy (J)	+0.645	[-0.874,+2.481]	0.5625

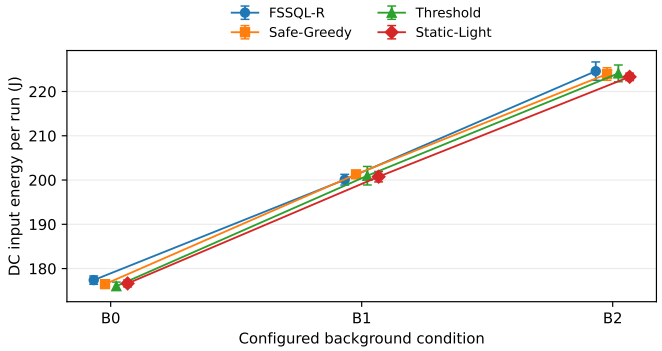


Fig. 2. Absolute Pi 4B DC input energy across the three configured background loads. Error bars show run-level SD over five seeds; no idle power is subtracted.

E. RQ4: Saved independent detector test

Table VII reports the fixed-detector evaluation completed on 11 September 2026, before the October implementation corrections. Those corrections neither change the saved detector/threshold artifacts nor reopen the official test. TinyDL has the highest saved F1 (0.8686), AUROC (0.9704), and PR-AUC (0.9784). FISVDD predicts all test rows positive and has AUROC 0.5351, indicating weak ranking as well as an unusable all-positive operating point. TinyDL and OI-SVDD+AS-ELM retain stronger ranking (AUROC 0.9704/0.9513) but substantial fixed-threshold FPR. These are distinct observations; we do not diagnose calibration error solely from AUROC or retune thresholds after test access.

TABLE VII
SAVED OFFICIAL UNSW-NB15 DETECTOR EVALUATION (82,332 ROWS; FIXED THRESHOLDS). THIS IS SEPARATE FROM THE DEVELOPMENT-WORKLOAD ONLINE COMPARISON.

Detector	F1	Precision	Recall	FPR
FISVDD	0.7102	0.5506	1.0000	1.0000
LUCID	0.8240	0.7020	0.9973	0.5186
TinyDL	0.8686	0.7802	0.9797	0.3382
OI-SVDD+AS-ELM	0.8521	0.7437	0.9975	0.4211

VII. DISCUSSION

A. Admissibility is not detection quality

The Pi 3B+ results establish that a calibrated guard can restrict the action set; the Pi 4B results show a nonbinding regime under the same nominal cap. Neither observation proves that shielding is universally necessary. In particular, a feasible FISVDD action can have unacceptable FPR. A resource shield constrains modeled latency and temperature, not classification quality. That distinction matters when the learned policy reaches an unvisited state and its tie-breaking rule selects a resource-feasible but weak detector.

B. Does the learning layer earn its cost?

Neither evaluated Pi regime establishes a practical need for the learned layer over the simpler same-guard rule. B0/B1 Pi 4B detection scores are close, while B2 and the Pi 3B+ binding comparison favor Safe-Greedy. FSSQL-R also incurs more switching. The two policies have different objectives as well as different decision horizons, so these comparisons are not a causal isolation of temporal learning. In addition, the zero-backlog experiments do not test the original queue-control motivation. The appropriate conclusion is limited to these reconstructed policies and workloads, not that RL can never help IDS scheduling.

The trace-supported B2 failure path suggests an evaluation requirement: report the fraction of online states supported by training, the action taken on an all-zero Q row, and that action's detection quality. Mean performance alone can hide these decisions. No after-test policy change is made to suppress the unfavorable B2 result. A support-aware default or a wider

training distribution would be a separate future study, not a retrospective correction to this experiment.

C. Sustainability and measurement boundaries

Recent sustainable edge-control studies act directly on GPU caps, sleep states, task placement, or DVFS [11]–[14]. Our action is detector selection under a fixed governor, and a normalized resource penalty is not an electrical energy measurement. Direct DC input measurement therefore tests rather than assumes energy savings. Here, background-load changes dominate the policy energy differences, and no consistent FSSQL-R advantage is established.

Adaptive edge inference and online IDS studies [16], [17], [24], [25] also motivate examining the complete deployed decision path. Our measured controller intervals exceed detector inference, and the Pi 3B+ outlier occurs despite substantial action-level guard headroom. Timing scopes, model residency, end-of-run work, and meter boundaries must therefore be stated explicitly. A timestamped file or an integrity hash alone does not show that the declared workload and controller were executed.

VIII. Limitations and Threats to Validity

The empirical scope is two actively cooled Pi-class configurations and UNSW-NB15; MCU applicability is prospective. There is no deployment near the 82°C cap and no accepted queue accumulation/recovery experiment. Joint empirical coverage is not a distribution-free guarantee outside the sampled hardware conditions. The two devices are not a single controlled cross-device benchmark: their workload, calibration, and training histories differ.

Pi 4B online inputs are reused scheduler-development records. The 576,000 record evaluations are not independent test samples, and five matched seeds per condition limit inference. Bootstrap intervals and unadjusted exact tests are descriptive, while the support analysis is post hoc. The load-process evidence establishes execution and configuration but not a separately measured CPU-utilization trajectory. ACK-bounded input energy includes tail work and is not inference-only energy.

The original experimental assets were unavailable, and the revision uses reconstructed implementations rather than an exact reproduction. The audit-invalid Pi 4B campaign is excluded, with corrections and replacement selection documented in the supplement. The official detector test predates the October corrections: it was not reopened, but the later scheduler implementation is not a fresh blinded test study. Saved confusion counts are recomputable; ranking metrics and the historical result-copy path have more limited retained provenance. These limitations constrain the claims and are not resolved by the release checksums.

IX. Conclusion

FSSQL-R combines calibrated feasible-action filtering with masked tabular Q-learning for multi-model IDS scheduling. Independent guard coverage and online execution provide evidence for device-specific admissibility, not unconditional end-to-end safety. The corrected Pi 4B experiment processes actual 16-record windows, uses the declared controllers and reward, and binds physical energy measurements to each run. It does not establish an incremental advantage for Q-learning: lower-load performance is close to Safe-Greedy, while high-load performance deteriorates through a documented unvisited-state/default-action path. Together with the Pi 3B+ comparison, these results favor testing simpler controllers before accepting the cost of learning, and reporting state support, false positives, and measured energy alongside average F1.

References

[1] C. Banbury, V. J. Reddi, P. Torelli, J. Holleman, N. Jeffries, C. Kiraly, P. Montino, D. Kanter, S. Ahmed, D. Pau *et al.*, "MLPerf tiny benchmark," *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

[2] V. J. Reddi, C. Cheng, D. Kanter, P. Mattson, G. Schmuelling, C.-J. Wu, B. Anderson, M. Breughe, M. Charlebois, W. Chou, R. Chukka, C. Coleman, S. Davis, P. Deng, G. Diamos, J. Duke, D. Fick, J. S. Gardner, I. Hubara, S. Idgunji, T. B. Jablin, J. Jiao, T. S. John, P. Kanwar, D. Lee, J. Liao, A. Lokhmotov, F. Massa, P. Meng, P. Micikevicius, C. Osborne, G. Pekhimenko, A. T. R. Rajan, D. Sequeira, A. Sirasao, F. Sun, H. Tang, M. Thomson, F. Wei, E. Wu, L. Xu, K. Yamada, B. Yu, G. Yuan, A. Zhong, P. Zhang, and Y. Zhou, "Mlperf inference benchmark," in *2020 ACM/IEEE 47th Annual International Symposium on Computer Architecture (ISCA)*, 2020, pp. 446–459.

[3] T. Tan and G. Cao, "Thermal-aware scheduling for deep learning on mobile devices with npu," *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 10 706–10 719, 2024.

[4] E. Altman, *Constrained Markov Decision Processes*, ser. Stochastic Modeling Series. Taylor & Francis, 1999.

[5] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, D. Precup and Y. W. Teh, Eds., vol. 70. PMLR, 06–11 Aug 2017, pp. 22–31.

[6] Y. Chow, O. Nachum, E. Duenez-Guzman, and M. Ghavamzadeh, "A lyapunov-based approach to safe reinforcement learning," in *Advances in Neural Information Processing Systems*, S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, Eds., vol. 31. Curran Associates, Inc., 2018.

[7] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, pp. 2669–2678, Apr. 2018.

[8] S. Carr, N. Jansen, S. Junges, and U. Topcu, "Safe reinforcement learning via shielding under partial observability," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 12, pp. 14 748–14 756, Jun. 2023.

[9] B. Könighofer, R. Bloem, N. Jansen, S. Junges, and S. Pranger, "Shields for safe reinforcement learning," *Commun. ACM*, vol. 68, no. 11, pp. 80–90, Oct. 2025.

[10] W. A. Malwi, F. Asiri, N. Alturki, N. Alnazzawi, D. Kasimatis, and N. Pitropakis, "Safety-aware reinforcement learning for self-healing intrusion detection in 5g-enabled iot networks," *Computers, Materials & Continua*, vol. 87, no. 2, p. 88, 2026.

[11] Y. Wang, M. Hao, H. He, W. Zhang, Q. Tang, X. Sun, and Z. Wang, "DRLCap: Runtime gpu frequency capping with deep reinforcement learning," *IEEE Transactions on Sustainable Computing*, vol. 9, no. 5, pp. 712–726, 2024.

[12] H. Ko, H. Choi, and S. Pack, "Restoration-aware sleep scheduling framework in energy harvesting internet of things: A deep reinforcement learning approach," *IEEE Transactions on Sustainable Computing*, vol. 10, no. 1, pp. 190–198, 2025.

[13] J. Gao, F. Zhou, M. Dong, L. Feng, P. Yu, K. Ota, and X. Qiu, "Optimal latency and energy-aware task scheduling in in-network computing paradigm: A deep reinforcement learning approach," *IEEE Transactions on Sustainable Computing*, vol. 10, no. 4, pp. 690–704, 2025.

[14] X. Li, T. Zhou, H. Wang, and M. Lin, "Energy-efficient computation with DVFS using deep reinforcement learning for multi-task systems in edge computing," *IEEE Transactions on Sustainable Computing*, vol. 10, no. 6, pp. 1116–1127, 2025.

[15] L. Xu, Z. Han, D. Zhao, X. Li, F. Yu, and C. Chen, "Addressing concept drift in IoT anomaly detection: Drift detection, interpretation, and adaptation," *IEEE Transactions on Sustainable Computing*, vol. 9, no. 6, pp. 913–924, 2024.

[16] X. Tan, H. Li, X. Xie, L. Guo, N. Ansari, X. Huang, L. Wang, Z. Xu, and Y. Liu, "Reinforcement learning based online request scheduling framework for workload-adaptive edge deep learning inference," *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 13 222–13 239, 2024.
[17] Z. Zhang, Y. Zhao, H. Li, C. Lin, and J. Liu, "DVFO: Learning-based DVFS for energy-efficient edge-cloud collaborative inference," *IEEE Transactions on Mobile Computing*, vol. 23, no. 10, pp. 9042–9059, 2024.
[18] D. M. Tax and R. P. Duin, "Support vector data description," *Machine Learning*, vol. 54, no. 1, pp. 45–66, Dec. 2004.
[19] G.-B. Huang, Q.-Y. Zhu, and C.-K. Siew, "Extreme learning machine: Theory and applications," *Neurocomputing*, vol. 70, no. 1, pp. 489–501, 2006, neural Networks.
[20] R. Doriguzzi-Corin, S. Millar, S. Scott-Hayward, J. Martínez-del Rincón, and D. Siracusa, "Lucid: A practical, lightweight deep learning solution for ddos attack detection," *IEEE Transactions on Network and Service Management*, vol. 17, no. 2, pp. 876–889, 2020.
[21] P. Fusco, G. P. Rimoli, and M. Ficco, "TinyIDS – an IoT intrusion detection system by tiny machine learning," in *Computational Science and Its Applications – ICCSA 2024 Workshops*. Cham: Springer Nature Switzerland, 2024, pp. 71–82.
[22] P. Warden and D. Situnayake, *TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers*. O'Reilly Media, 2019.
[23] J. Lin, W.-M. Chen, Y. Lin, j. cohn, C. Gan, and S. Han, "MCUNet: Tiny deep learning on IoT devices," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33. Curran Associates, Inc., 2020, pp. 11 711–11 722.
[24] M. Nakip and E. Gelenbe, "Online self-supervised deep learning for intrusion detection systems," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 5668–5683, 2024.
[25] X. Yang, F. Tong, F. Jiang, and G. Cheng, "A lightweight and dynamic open-set intrusion detection for industrial internet of things," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 2930–2943, 2025.
[26] J. García and F. Fernández, "A comprehensive survey on safe reinforcement learning," *Journal of Machine Learning Research*, vol. 16, no. 42, pp. 1437–1480, 2015.
[27] N. Moustafa and J. Slay, "UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *2015 Military Communications and Information Systems Conference (MilCIS)*, 2015, pp. 1–6.

Nuonan Ouyang is a Ph.D. candidate at James Cook University, Australia. His research focuses on lightweight edge/IoT cybersecurity, on-device intrusion detection, and deployable learning-based control under resource constraints.

Adrian Shatte is a Senior Lecturer in Information Technology at James Cook University, Australia. His research interests include data science, intelligent systems, digital wellbeing, and applied machine learning.

Zhigang Lu is a Senior Lecturer with Western Sydney University, Australia. His research interests include privacy, security, robustness, and trustworthy AI/ML systems.

Chao Chen is an Associate Professor at RMIT University, Australia. His research spans AI for cybersecurity, data analytics, and security of AI systems.

Wei Xiang is a La Trobe Distinguished Professor and Cisco Research Chair of AI and IoT at La Trobe University, Australia. His research spans communications, IoT, and applied AI.